Active-learning combination catalogues for MALDI-TOF resistance models

Catalogues AL-A to AL-E · search for a better way to choose which isolates the lab tests · 5 October 2026

In one paragraph. E1–E4 showed three things: (1) no single selection rule beats random on model quality; (2) testing the uncertain isolates cuts missed resistances by 33–73%; (3) the threshold and conformal bounds set on 2015–16 data break under drift (sensitivity 73–80% instead of 90%). So the goal of the search is a combination that **keeps random's model quality, keeps uncertainty's safety, and repairs calibration**. We split each method into 5 slots, listed every option per slot (AL-A), removed impossible or pointless combinations (AL-B: 2 430 → 1 424 valid → 559 robust and affordable), ranked them with evidence-based scores (AL-C), checked the literature for novelty (AL-D) and chose 16 to **actually run** on a development split that never touches 2018 (AL-E). The winners, chosen by a rule fixed in advance, are then run once on the real 2017 → 2018 test.

Terms used throughout

Words used here

- **AUROC** — how well the model ranks resistant above susceptible isolates (1 = perfect, 0.5 = guessing).
- **VME (very major error)** — a resistant isolate reported as susceptible: the dangerous mistake.
- **missed resistances** — VMEs among the isolates NOT sent to the lab during 2017, i.e. patients who got the model's wrong answer.
- **threshold t** — the model calls 'resistant' when its probability p ≥ t; t is chosen so that 90% of resistant isolates are caught (sensitivity 0.90).
- **conformal bounds** — per-class cut-offs that turn p into a set of plausible labels {S}, {R} or {S, R}; they promise the true label is in the set ≥ 90% of the time, if new data look like the calibration data.
- **calibration** — choosing t and the conformal bounds from labelled isolates.
- **random share** — the part of each round's budget bought uniformly at random; it is an unbiased sample of the new isolates.
- **prequential** — 'predict, then learn': an isolate is scored by the model in use before that isolate is added to training, so its score is out-of-sample.
- **development split** — train 2015–16, buy labels from 2017 batches 1–2, test on 2017 batches 3–4. Used for the search so that 2018 stays untouched.

AL-A · Components (5 slots)

Scores are 0–3. **MQ** model quality on drifted data · **SF** safety (fewer missed resistances) · **CV** calibration validity · **CC** compute cheapness · **RB** robustness (low risk of failure). Evidence: **E** = our E1–E4 results, **L** = literature (tier A/B/C), **P** = prediction from the mechanism.

ID	Selection rule (targeted share)	MQ	SF	CC	RB	Evidence
S1	random	2	1	3	3	E: best model quality in E1 (no strategy beat it)
S2	uncertainty (|p - t| smallest)	1	3	3	2	E: missed resistances -33/-73/-51% vs random; AUROC worse for K. pneumoniae
S3	diversity (k-center)	2	1.5	2	3	E: = random on AUROC; missed resistances slightly lower (n.s.)
S4	hybrid (uncertainty then clusters)	2	1.5	2	2	E: = random; best gap closure for E. coli at 800 (n.s.)
S5	resistance hunting (highest p)	1	0.5	3	2	P: buys isolates already called R, so it cannot prevent missed resistances
S6	VME-risk (called S, closest to threshold)	1	3	3	2	P: spends the whole budget where a missed resistance can happen (asymmetric uncertainty); L: cost-sensitive AL (Krishnamurthy et al., ICML 2017, A)
S7	drift novelty (far from training data)	2	1.5	2	2	P: covers the new regions of spectrum space created by drift
S8	committee disagreement (3 bootstrap models)	2	2	1	2	L: query-by-committee; 3x training cost
S9	conformal ambiguity (routing)	1.5	1	3	1	E: with old calibration it missed MORE resistances than random (coverage broke)

ID	Budget split	Effect in the scoring	Evidence
B1	one rule (100% targeted)	profile of the rule	
B2	50% targeted + 50% random	MQ = max(rule, random); SF = rule – 0.5 (≥ random)	
B3	90% targeted + 10% random	MQ = max(rule, random) – 0.25 if the rule is weaker; SF = rule – 0.1	L: 10% randomness rescues uncertainty sampling (Lu, Li, Lin, ICML 2023 workshop, B)
B4	schedule: rounds 1-2 random, rounds 3-4 targeted	average of rule and random, + 0.25	
B5	bandit picks targeted or random each round	MQ max – 0.25; SF average; RB – 1; CC 2	L: ALBL (Hsu & Lin, AAAI 2015, A)

ID	Retraining	Effect in the scoring	Evidence
T1	old and new labels weighted equally	—	
T2	new labels weighted 3x (recency)	MQ +0.25; RB -0.25	P: resistance mechanisms drift; recent isolates more relevant
T3	drop the oldest year (2015)	—	E: 2015 holds only 47-97 isolates per pair, so no effect expected
T4	importance-weight the targeted labels	MQ +0.25; RB -0.25	L: importance-weighted AL removes sampling bias (Beygelzimer et al., ICML 2009, A)

ID	Calibration (threshold + conformal bounds)	Effect in the scoring	Evidence

C1	old 2015-16 validation only	CV 0	E: sensitivity 73-80% vs 90% target; conformal miscoverage >= 14-19% vs 10%
C2	25% of bought labels held out (E5)	CV 2	running now (E5)
C3a	random share held out, calibration only	CV 3	P: an unbiased sample of the new distribution, so the 90% target and conformal guarantee hold again
C3b	random share, scored before retraining (prequential)	CV 2.5	P: same unbiased sample, used for training too; scores come from the previous model (slight mismatch)
C4	weighted conformal (density ratio from unlabelled 2017 spectra)	CV 2; CC 2	L: conformal under covariate shift (Tibshirani et al., NeurIPS 2019, A); covers covariate shift only
C5	adaptive conformal / threshold tracking on lab feedback	CV 2.5	L: ACI (Gibbs & Candes, NeurIPS 2021, A)

ID	Free unlabelled spectra	Effect in the scoring	Evidence
U0	none	—	
U1	pseudo-labels with cut-offs tied to the threshold	SF -0.5; RB -1	E: E2 (flawed cut-offs) hurt badly
U2	conformal-certified pseudo-labels (recalibrated sets, alpha 0.05)	MQ +0.25; RB -0.5	P: only isolates with a single label in a valid prediction set

AL-B · Compatibility rules

1. Random + random share (S1 with B2–B5) is the same as random → removed.
2. Conformal routing (S9) with the old calibration (C1) → removed (measured failure in E3).
3. Re-calibration on the random share (C3a, C3b) and importance weights (T4) need a random share (B2, B3, B5).
4. Conformal pseudo-labels (U2) need prediction sets re-calibrated on new labels (not C1 or C4).
5. The bandit (B5) needs fresh labelled feedback (not C1).
6. Dropping 2015 (T3) equals T1 here (2015 holds only 47–97 isolates per pair) → merged.
7. Calibration fed by bought labels (C2, C5) is biased by the targeted share: CV capped at 1.5 (100% targeted), 1.6 (90%), 2.0 (50%).
8. Held-out shares cost training labels: MQ −0.25 (C2), −0.75 (C3a with 50% random), −0.1 (C3a with 10%).

Stage	Count
All slot combinations (9 × 5 × 3 × 6 × 3)	2430
Valid after rules 1–6	1424
Robust (RB ≥ 1.5) and affordable (CC ≥ 1)	559

AL-C · Ranking

$$Q = 100 \cdot (0.35 \cdot MQ + 0.35 \cdot SF + 0.30 \cdot CV) / 3 \quad (1)$$

where

MQ, SF, CV combination scores from the slot rules above (0–3)

NV novelty in MALDI-AMR: 1 = active learning for isolate selection (not found in AMR), +1 calibration-under-drift part, +1 a principle not found even in general ML (random-share re-calibration or VME-risk selection)

Top 24 of 559. Many rows differ only in retraining or pseudo-labels; those are tested as single changes in the shortlist rather than as separate winners.

#	Combination	In words	MQ	SF	CV	RB	Q	NV
1	**S2-B2-T2-C3b-U0**	uncertainty (|p - t| smallest) · 50% targeted + 50% random · new labels weighted 3x (recency) · random share, scored before retraining (prequential)	2.25	2.5	2.5	1.75	**80.4**	3
2	**S2-B2-T4-C3b-U0**	uncertainty (|p - t| smallest) · 50% targeted + 50% random · importance-weight the targeted labels · random share, scored before retraining (prequential)	2.25	2.5	2.5	1.75	**80.4**	3
3	**S6-B2-T2-C3b-U0**	VME-risk (called S, closest to threshold) · 50% targeted + 50% random · new labels weighted 3x (recency) · random share, scored before retraining (prequential)	2.25	2.5	2.5	1.75	**80.4**	3
4	**S6-B2-T4-C3b-U0**	VME-risk (called S, closest to threshold) · 50% targeted + 50% random · importance-weight the targeted labels · random share, scored before retraining (prequential)	2.25	2.5	2.5	1.75	**80.4**	3
5	**S2-B2-T1-C3b-U2**	uncertainty (|p - t| smallest) · 50% targeted + 50% random · old and new labels weighted equally · random share, scored before retraining (prequential) · conformal-certified pseudo-labels (recalibrated sets, alpha 0.05)	2.25	2.5	2.5	1.5	**80.4**	3
6	**S6-B2-T1-C3b-U2**	VME-risk (called S, closest to threshold) · 50% targeted + 50% random · old and new labels weighted equally · random share, scored before retraining (prequential) · conformal-certified pseudo-labels (recalibrated sets, alpha 0.05)	2.25	2.5	2.5	1.5	**80.4**	3

7	**S2-B2-T1-C3b-U0**	uncertainty (|p - t| smallest) · 50% targeted + 50% random · old and new labels weighted equally · random share, scored before retraining (prequential)	2	2.5	2.5	2	**77.5**	3
8	**S6-B2-T1-C3b-U0**	VME-risk (called S, closest to threshold) · 50% targeted + 50% random · old and new labels weighted equally · random share, scored before retraining (prequential)	2	2.5	2.5	2	**77.5**	3
9	**S6-B3-T2-C4-U0**	VME-risk (called S, closest to threshold) · 90% targeted + 10% random · new labels weighted 3x (recency) · weighted conformal (density ratio from unlabelled 2017 spectra)	2	2.9	2	1.75	**77.2**	3
10	**S6-B3-T4-C4-U0**	VME-risk (called S, closest to threshold) · 90% targeted + 10% random · importance-weight the targeted labels · weighted conformal (density ratio from unlabelled 2017 spectra)	2	2.9	2	1.75	**77.2**	3
11	**S2-B3-T2-C4-U0**	uncertainty (|p - t| smallest) · 90% targeted + 10% random · new labels weighted 3x (recency) · weighted conformal (density ratio from unlabelled 2017 spectra)	2	2.9	2	1.75	**77.2**	2
12	**S2-B3-T4-C4-U0**	uncertainty (|p - t| smallest) · 90% targeted + 10% random · importance-weight the targeted labels · weighted conformal (density ratio from unlabelled 2017 spectra)	2	2.9	2	1.75	**77.2**	2
13	**S2-B2-T2-C3a-U0**	uncertainty (|p - t| smallest) · 50% targeted + 50% random · new labels weighted 3x (recency) · random share held out, calibration only	1.5	2.5	3	1.75	**76.7**	3
14	**S2-B2-T4-C3a-U0**	uncertainty (|p - t| smallest) · 50% targeted + 50% random · importance-weight the targeted labels · random share held out, calibration only	1.5	2.5	3	1.75	**76.7**	3
15	**S6-B2-T2-C3a-U0**	VME-risk (called S, closest to threshold) · 50% targeted + 50% random · new labels weighted 3x (recency) · random share held out, calibration only	1.5	2.5	3	1.75	**76.7**	3
16	**S6-B2-T4-C3a-U0**	VME-risk (called S, closest to threshold) · 50% targeted + 50% random · importance-weight the targeted labels · random share held out, calibration only	1.5	2.5	3	1.75	**76.7**	3
17	**S2-B2-T1-C3a-U2**	uncertainty (|p - t| smallest) · 50% targeted + 50% random · old and new labels weighted equally · random share held out, calibration only · conformal-certified pseudo-labels (recalibrated sets, alpha 0.05)	1.5	2.5	3	1.5	**76.7**	3
18	**S6-B2-T1-C3a-U2**	VME-risk (called S, closest to threshold) · 50% targeted + 50% random · old and new labels weighted equally · random share held out, calibration only · conformal-certified pseudo-labels (recalibrated sets, alpha 0.05)	1.5	2.5	3	1.5	**76.7**	3
19	**S6-B2-T2-C5-U0**	VME-risk (called S, closest to threshold) · 50% targeted + 50% random · new labels weighted 3x (recency) · adaptive conformal / threshold tracking on lab feedback	2.25	2.5	2	1.75	**75.4**	3
20	**S6-B2-T4-C5-U0**	VME-risk (called S, closest to threshold) · 50% targeted + 50% random · importance-weight the targeted labels · adaptive conformal / threshold tracking on lab feedback	2.25	2.5	2	1.75	**75.4**	3
21	**S6-B2-T2-C4-U0**	VME-risk (called S, closest to threshold) · 50% targeted + 50% random · new labels weighted 3x (recency) · weighted conformal (density ratio from unlabelled 2017 spectra)	2.25	2.5	2	1.75	**75.4**	3

22	**S6-B2-T4-C4-U0**	VME-risk (called S, closest to threshold) · 50% targeted + 50% random · importance-weight the targeted labels · weighted conformal (density ratio from unlabelled 2017 spectra)	2.25	2.5	2	1.75	**75.4**	3
23	**S6-B2-T1-C5-U2**	VME-risk (called S, closest to threshold) · 50% targeted + 50% random · old and new labels weighted equally · adaptive conformal / threshold tracking on lab feedback · conformal-certified pseudo-labels (recalibrated sets, alpha 0.05)	2.25	2.5	2	1.5	**75.4**	3
24	**S2-B2-T2-C5-U0**	uncertainty (|p - t| smallest) · 50% targeted + 50% random · new labels weighted 3x (recency) · adaptive conformal / threshold tracking on lab feedback	2.25	2.5	2	1.75	**75.4**	2

These are **analytic** scores, used only to choose what to run. Unlike the world-model catalogues, here every shortlisted combination is actually run (each job takes ~30 s), so the final choice comes from measurements, not from these numbers.

AL-D · Literature and novelty check

Idea	Closest work found	Tier	What it means for us
Active learning to choose which isolates to phenotype for MALDI-TOF resistance models	not found (searched 5 Oct 2026; formal Scopus/PubMed check pending)	—	new in AMR
MALDI-TOF resistance prediction on DRIAMS, temporal decay	Weis et al., Nature Medicine 2022	**A**	motivation only
Conformal prediction for MALDI-TOF resistance	Corvelo Benz et al., RECOMB 2025 (LNCS)	**A**	published: routing alone is not new
10% randomness rescues uncertainty sampling	Lu, Li, Lin, ICML 2023 workshop	**B**	general ML, no calibration use
Active learning + selective prediction under shift	Chen et al., ASPEST, TMLR 2024	**A**	general ML; ensembles + self-training
Bandit choice among AL strategies (ALBL)	Hsu & Lin, AAAI 2015	**A**	general ML
Importance-weighted active learning	Beygelzimer et al., ICML 2009	**A**	general ML
Cost-sensitive active learning	Krishnamurthy et al., ICML 2017	**A**	general ML; not VME-targeted
Conformal prediction under covariate shift (weighted)	Tibshirani et al., NeurIPS 2019	**A**	general ML
Adaptive conformal inference under shift	Gibbs & Candès, NeurIPS 2021	**A**	general ML
Active learning under drift (malware), label-efficient updates	Chen, Ding, Wagner, USENIX Security 2023	**A**	other domain; no calibration share
Conformal evaluation under drift (malware)	Barbero et al., Transcending, IEEE S&P 2022	**A**	other domain
Selective labelling with FDR guarantee (AI vs human labels)	arXiv 2510.14581, 2025	**C**	related idea, preprint
Calibrated uncertainty sampling	Bui et al., arXiv 2510.03162, 2025	**C**	calibration of the sampler, not of the decision threshold
Split budget: random share re-sets threshold + conformal bounds (scored before retraining), targeted share for safety	not found in general ML or AMR	—	the candidate contribution
Selection aimed at missed resistances only (VME-risk)	not found for AMR; cost-sensitive AL exists in general	—	new in AMR

Novelty, honestly. The individual pieces (mixing random into uncertainty sampling, conformal prediction, weighted/adaptive conformal, cost-sensitive AL) all exist. Not found anywhere: **(1)** active learning to choose which isolates to phenotype for a drifting MALDI-TOF resistance model; **(2)** using the random share of the budget, scored before retraining, to re-set the decision threshold and conformal bounds under drift while the targeted share protects patients; **(3)** selection aimed only at the isolates where a missed resistance can occur (VME-risk). Claims (2) and (3) must still be checked in Scopus/PubMed via the library before writing "first".

AL-E · Shortlist for the development search (D01–D16)

Built as chains of single changes from the anchors, so every part's effect can be read off directly (an ablation by design). Bounds (no retraining, full retraining) are also run. 16 combinations × 3 pairs × 5 seeds × 2 budgets + bounds = **510 jobs** (~3–4 GPU-hours).

Code	Parts	What it isolates	Prediction vs random (D01)	Confidence
D01	random, old calibration	anchor = E1 random	—	
D02	uncertainty, old calibration	anchor = E1 uncertainty	safety ↑↑, AUROC ↓ (seen in E1)	high

D03	random, 25% held out (E5)	anchor = E5	calibration ↑, AUROC ↓ slightly	medium
D04	VME-risk alone	the new selection rule	safety ↑↑ (≥ D02), AUROC ↓	medium
D05	uncertainty 50% + random 50%	mixing alone	AUROC ≈ random, safety ↑	medium
D06	D05 + prequential re-calibration	the calibration step	AUROC ≈ random, safety ↑, sensitivity back near 90%	medium
D07	VME-risk 50% + random 50% + prequential	core with VME-risk	as D06 with larger safety gain	medium-low
D08	D06 but random share held out	unbiased but costs labels	best calibration, AUROC ↓	medium
D09	D06 + new labels ×3	recency	AUROC +0 to +0.01	low
D10	D06 + importance weights	bias correction	AUROC ≈ D06	low
D11	D06 + conformal pseudo-labels	fixed pseudo-labelling	AUROC +0 to +0.01; risk of VME rise	low
D12	D07 + new labels ×3	VME-risk + recency	as D07	low
D13	VME-risk 90% + random 10% + weighted conformal	label-free calibration	safety ↑↑, calibration partly fixed	low
D14	VME-risk 50% + random 50% + adaptive (ACI)	online calibration	calibration ↑ if feedback suffices	low
D15	hybrid 50% + random 50% + prequential	diversity-flavoured core	AUROC ≈ random	medium-low
D16	drift-novelty 50% + random 50% + prequential	covering new regions	AUROC ≈ random or ↑	low

The leading candidate (working name SPARC)

Split-budget, **P**requential, **A**ctive, **R**e-**C**alibrated selection = D06 / D07 family. Each round, half of the budget is bought at random and half is targeted; the random half is scored by the current model *before* it is trained on, and those scores re-set the threshold and the conformal bounds of the retrained model. (The name is provisional: SPARC is a common acronym; check before use.)

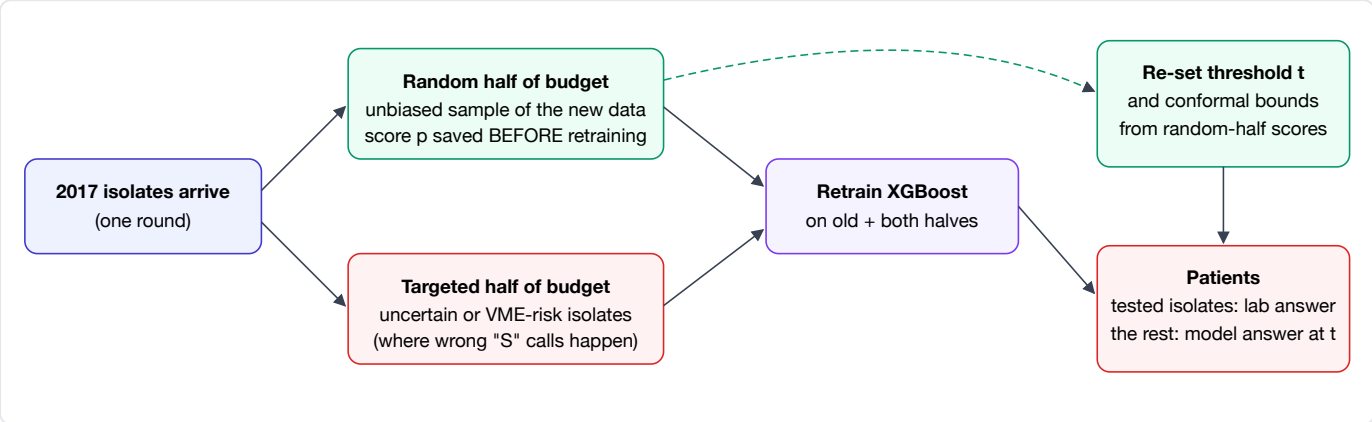


Figure 1. One round of the candidate. Green = the random half (learning + calibration), red = the targeted half (learning + patient safety).

$$t_r = \text{largest } t \text{ such that } \#\{\{\, i \in \mathcal{R}_{\le r},\, y_i = R : p_i^{(acq)} \ge t \,\}\} \ge 0.90 \cdot \#\{\{\, i \in \mathcal{R}_{\le r},\, y_i = R \,\}\} \tag{2}$$

where

- $\mathcal{R}_{\le r}$ the random-share isolates bought up to round r (with the 2015–16 validation set added while they hold fewer than 10 resistant isolates)
- $p_i^{(acq)}$ the probability the model in use gave isolate i when it arrived, i.e. before it was trained on i
- 0.90 the target sensitivity

$$q_c = \text{the } \lceil (n_c+1)(1-\alpha) \rceil\text{-th smallest of } \{\, 1 - p_i(c) : i \in \mathcal{R}_{\le r},\, y_i = c \,\} \tag{3}$$

where

- q_c conformal bound for class c (S or R); label c is in the set if $1 - p(c) \le q_c$
- α 0.10, so each class's true label is in the set ≥ 90% of the time if the random share looks like future isolates
- n_c number of random-share isolates of class c

Why it could work. The random share is an unbiased sample of the drifted data, so eq. 2 and 3 are calibrated on the right distribution (the reason E1's threshold failed). Because its scores are out-of-sample, no labels are wasted on a separate hold-out (unlike E5/D08). The targeted half keeps most of uncertainty sampling's safety gain, and the random half keeps random's model quality (randomness rescues uncertainty sampling, Lu et al. 2023).

Why it could fail. The threshold comes from the previous model's scores but is applied to the retrained model (small mismatch); 50 random labels per round hold only ~10 resistant isolates, so early thresholds are noisy; isolates arrive in pseudo-batches (no acquisition dates), so within-2017 drift cannot be tested.

Protocol (fixed before any result, also in RESEARCH_LOG)

1. **Development search** (D1_dev, 510 jobs): rank by the rule below.
2. **Rule:** (a) AUROC non-inferior to D01: mean difference over the 6 pair × budget cells ≥ −0.005 and no pair significantly worse (Wilcoxon over seeds, Holm $p < 0.05$); (b) among those, rank by missed resistances relative to D01 and by |sensitivity − 0.90| on the development test, with R-class conformal coverage ≥ 0.85; sum the two ranks; ties → fewer changed parts.
3. **Freeze** the top 2 and the chain of single changes leading to them in RESEARCH_LOG (commit = timestamp).

4. **Confirm once** on 2017 → 2018 (seeds 0–9, budgets 100/400/800), compared by seed with E1 random/uncertainty and E5.
5. **Success** = on 2018, AUROC within 0.005 of random, missed resistances significantly lower than random, and sensitivity closer to 0.90 than E1. Anything else is reported as found.

References

Weis et al., Nature Medicine 28, 2022 **A** Corvelo Benz et al., RECOMB 2025, LNCS **A** Chen et al., ASPEST, TMLR 2024 **A** Lu, Li, Lin, ICML 2023 workshop (AI & HCI) **B** Hsu & Lin, ALBL, AAAI 2015 **A** Beygelzimer, Dasgupta, Langford, ICML 2009 **A** Krishnamurthy et al., ICML 2017 **A** Tibshirani et al., NeurIPS 2019 **A** Gibbs & Candès, NeurIPS 2021 **A** Chen, Ding, Wagner, USENIX Security 2023 **A** Barbero et al., IEEE S&P 2022 **A** Sener & Savarese, ICLR 2018 (k-center) **A** arXiv 2510.14581 (2025) **C** arXiv 2510.03162 (2025) **C**